

LARGE ERDŐS–STRAUS WITNESSES: Ω-RESULTS FOR THE MULTIPLIER AND SLICE PARAMETERS

ANONYMOUS

ABSTRACT. For a prime p let $W(p)$ be the least modulus $M \equiv 3 \pmod{4}$ of a “multiplier” congruence that forces a solution of $4/p = 1/x + 1/y + 1/z$, and let $ck_{\min}(p)$ be the least product ck of a positive Type I slice. Both are expected to be at most a power of $\log p$ for almost all p , and are known to exceed a constant times $\log p$ for infinitely many p . We prove that $W(p) \geq (\log p)^2 \exp(-C \log \log p / \log \log \log p)$ for infinitely many primes p in Mordell’s hard classes. The construction is a prime-local sieve: the class of one is imposed modulo all small primes, a congruence saving bounds the residual sieve mass by $T^{o(1)}$, and the sieve is transferred to primes by the Linnik-range prime number theorem of Thorner and Zaman, whose relative error term absorbs a possible Siegel zero. We show that exponent 2 is the limit of every prime-local design and isolate the exact combinatorial input needed to go further. For the slice parameter we prove that congruence methods certify exactly the least quadratic non-residue, giving $ck_{\min}(p) \gg \log p \log \log \log p$ infinitely often (via Lau–Wu, after Graham–Ringrose) and showing that a power saving would be a breakthrough on the least non-residue. On the Haar side we prove that the avoider density satisfies $\log(1/\delta^*(T)) \leq T^{1/3+o(1)}$.

1. INTRODUCTION

The Erdős–Straus conjecture asserts that $4/n = 1/x + 1/y + 1/z$ is solvable in positive integers for every $n \geq 2$. It suffices to treat primes, it has been verified for $n \leq 10^{18}$ ([10], §6; [11] for 10^{17}), and Mordell’s identities settle every prime outside the six classes $p \equiv 1^2, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$ [9]; we call such primes *hard*. On the exceptional-set side, Vaughan [13] proved that the number $E(N)$ of $n \leq N$ without a representation satisfies $E(N) \ll N \exp\{-c(\log N)^{2/3}\}$; see [10] for a uniform modern account and for many exceptions to the analogous statement with large numerators, and [4] for the average number of solutions, which over primes $p \leq N$ is $(\log N)^3$ up to a factor $\log \log N$.

This note is about the *pointwise size* of the smallest witness. Two natural witness parameters are the following.

The multiplier parameter. For $M \equiv 3 \pmod{4}$ put $A_M = (M + 1)/4$. If $uvw = A_M$ and $n \equiv -uv^{-1} \pmod{M}$, then $s = (nv + u)/M$ is a positive

integer and

$$\frac{4}{n} = \frac{1}{suw} + \frac{1}{nsuv} + \frac{1}{nuvw} \quad (1)$$

(direct verification; this is the classical multiplier family, cf. [13, 4]). Define

$$W(n) = \min\{M \equiv 3 \pmod{4} : n \equiv -uv^{-1} \pmod{M} \text{ for some } uvw = A_M\}, \quad (2)$$

with $W(n) = +\infty$ if no such M exists. Thus $W(p) \leq T$ supplies an explicit representation with a modulus at most T .

The slice parameter. A Type I solution with parameters (a, b, c, k) satisfies $p(a+b) = k(4abc-1)$ and gives $4/p = 1/(ack) + 1/(bck) + 1/(pabc)$. For an admissible pair (c, k) (see §8) let $M_{c,k}(p)$ be the number of divisors of $p^2 + 4ck^2$ lying in the class $-p \pmod{4ck}$; the slice is positive iff $M_{c,k}(p) > 0$. Put $ck_{\min}(p) = \min\{ck : M_{c,k}(p) > 0, \text{sf}(c) \notin \{1, 2, 3, 6\}\}$ (the cores 1, 2, 3, 6 are excluded because they are deterministic on hard primes).

Both parameters are expected to be small for almost all primes. For W , tails of the form $\#\{p \leq N : W(p) > T\} \ll N \exp\{-c(\log T)^2 \log \log T\}$ (in the range $\log N \gg (\log T)^3 \log \log T$) are obtained in companion work whose assembly is still under internal review; we do not use them. In the other direction, the class of one (every prime $p \equiv 1 \pmod{\text{lcm}\{M \leq T\}}$) has $W(p) > T$ together with Linnik's theorem shows $W(p) \gg \log p$ infinitely often; with Chang's least-prime bound for smooth moduli [2] the constant becomes $5/8$. The same is true for $ck_{\min}$ via genus characters. Whether $W(p)/\log p$ is unbounded was open.

Theorem 1.1 (Proved modulo Theorem 4.1; effective). *There is an absolute constant C such that for infinitely many hard primes p*

$$W(p) \geq (\log p)^2 \exp\left(-C \frac{\log \log p}{\log \log \log p}\right).$$

More precisely, for every sufficiently large T there is a prime $p \equiv 1 \pmod{840}$ with $W(p) > T$ and $\log p \leq T^{1/2} \exp(O(\log T / \log \log T))$.

In particular the statement “ $W(p) \leq (\log p)^A$ for all large p ” is false for every $A < 2$. The cited input (Theorem 4.1) is Thorner and Zaman's prime number theorem in the Linnik range [12], including the McCurley zero-free region statement quoted there. No hypothesis on Siegel zeros is needed: the exceptional zero is absorbed by a uniqueness lemma across the family of moduli (Lemma 4.2) and a two-case analysis (Theorem 5.1).

The exponent 2 is the natural limit of the method (Proposition 7.2); beyond it one needs a pointwise “hypergraph” minorant, and Theorem 7.4 shows that such a minorant would suffice, the prime side being completely handled.

For the slice parameter the picture is different: the only congruence mechanism is the genus character, and it certifies exactly the least quadratic non-residue n_p .

Theorem 1.2. (a) (Proved.) $ck_{\min}(p) \geq n_p$ for every prime $p \equiv 1 \pmod{24}$, and every fixed-period congruence minorant for $\{ck_{\min}(p) > T\}$ is supported on primes with $n_p > T$ (Corollary 8.5).
 (b) (Proved modulo [7, Prop. 5.1].) $ck_{\min}(p) \gg \log p \cdot \log \log \log p$ for infinitely many hard primes.

Thus a congruence proof of $ck_{\min}(p) > (\log p)^{1+\delta}$ infinitely often would improve the best known unconditional (Graham–Ringrose) and GRH-conditional (Montgomery) Ω -results for n_p , and would contradict the usual random model for n_p .

Finally, on the Haar (profinite) side, let $\delta^*(T)$ be the Haar measure of the set of $n \in \widehat{\mathbb{Z}}^\times$ with $n \equiv 1 \pmod{24}$ and $W(n) > T$, normalised within the class $1 \pmod{24}$. The class of one gives $\log(1/\delta^*(T)) \leq (2/3 + o(1))T$, and a crude prime-local bound gives $T^{1/2+o(1)}$.

Theorem 1.3 (Proved). $\log(1/\delta^*(T)) \leq T^{1/3+o(1)}$.

Numerically $\log(1/\delta^*(T))$ grows like a small power of $\log T$, which suggests that much larger values of $W(p)$ occur. We do not pursue heuristics here.

Status conventions. Every theorem header carries a label. *Proved* means proved in full here. *Proved modulo X* means proved here assuming the cited statement X, whose statement we checked in the source but whose proof we did not check. *Cited* marks external statements.

2. THE ATOMS OF THE MULTIPLIER PROBLEM

For $M \equiv 3 \pmod{4}$ put $\mathcal{R}(M) = \{-4D \bmod M : D \mid A_M^2\}$.

Lemma 2.1 (Proved). $\{-uv^{-1} \bmod M : uvw = A_M\} = \mathcal{R}(M)$. Consequently, for every integer n ,

$$W(n) > T \iff n \bmod M \notin \mathcal{R}(M) \text{ for every } M \leq T, \ M \equiv 3 \pmod{4}.$$

Proof. Since $4A_M \equiv 1 \pmod{M}$ and $\gcd(A_M, M) = 1$, from $uvw = A_M$ we get $v^{-1} \equiv 4uw$, hence $-uv^{-1} \equiv -4u^2w$ and $D = u^2w$ divides A_M^2 . Conversely let $D \mid A_M^2$. For each prime q with $q^a \parallel A_M$ and $q^d \parallel D$ ($d \leq 2a$) put $u_q = \max(0, d-a)$, $w_q = d - 2u_q$, $v_q = a - u_q - w_q$; these are non-negative, so $u = \prod q^{u_q}$, $v = \prod q^{v_q}$, $w = \prod q^{w_q}$ satisfy $uvw = A_M$ and $u^2w = D$. $\square$

Lemma 2.2 (Class of one; proved). $1 \notin \mathcal{R}(M)$ for every $M \equiv 3 \pmod{4}$.

Proof. If $-4D \equiv 1$ then $D \equiv -A_M \pmod{M}$. The map $D \mapsto A_M^2/D$ preserves this congruence, so we may assume $D \leq A_M$; then $0 < D + A_M \leq 2A_M < M$ while $M \mid D + A_M$. $\square$

3. THE PRIME-LOCAL RESIDUAL SYSTEM

Fix T large and put $\mathcal{L} = \log T$,

$$y = \sqrt{T} \exp(3\mathcal{L}/\log \mathcal{L}),$$

$$Q = \text{lcm}(24, \ell^{e_\ell} : \ell \leq y \text{ prime}, \ell^{e_\ell} \leq T \text{ maximal}),$$

and let $\mathcal{U}$ be the set of primes in $(y, T]$. Then $\log Q \leq 2y$ for large T (only $\ell \leq \sqrt{T}$ have $e_\ell \geq 2$), and $840 \mid Q$. For $\ell \in \mathcal{U}$ let

$$F_\ell = \{-4D \bmod \ell : m \leq T/\ell, m\ell \equiv 3 \pmod{4}, D \mid A_{m\ell}^2, m \mid 4D+1\},$$

$$g(\ell) = \frac{|F_\ell|}{\ell-1}, \quad S = \sum_{\ell \in \mathcal{U}} g(\ell).$$

All elements of F_ℓ are units.

Lemma 3.1 (Prime-local reduction; proved). *If $n \equiv 1 \pmod{Q}$ and $n \bmod \ell \notin F_\ell$ for all $\ell \in \mathcal{U}$, then $W(n) > T$.*

Proof. Let $M \leq T$, $M \equiv 3 \pmod{4}$. If every prime factor of M is at most y , then $M \mid Q$, so $n \equiv 1 \pmod{M}$ and Lemma 2.2 applies. Otherwise let $\ell > y$ divide M ; since $\ell^2 > T$ we have $\ell \parallel M$, and $m = M/\ell < T/y \leq y$, so $m \mid Q$ and $n \equiv 1 \pmod{m}$. If $n \equiv -4D \pmod{M}$ with $D \mid A_M^2$, then $m \mid 4D+1$ and $n \bmod \ell \in F_\ell$. Now use Lemma 2.1. $\square$

Lemma 3.2 (Local sizes; proved). *$g(\ell) \leq 2T\tau^*(T)^2/y^2$, where $\tau^*(T) = \max_{n \leq T} \tau(n)$. In particular $\max_\ell g(\ell) \leq 1/16$ for large T .*

Proof. For each $m \leq T/\ell$ there are at most $\tau(A^2) \leq \tau(A)^2$ choices of D , and $1/(\ell-1) \leq 2/\ell$. Since $\log \tau^*(T) \leq (\log 2 + o(1))\mathcal{L}/\log \mathcal{L}$ [6, Thm. 317], the bound is $\leq 2 \exp((2 \log 2 - 6 + o(1))\mathcal{L}/\log \mathcal{L})$. $\square$

The next lemma is the key arithmetic input: the congruence $m \mid 4D+1$ saves a factor m .

Lemma 3.3 (Mass bound; proved). *With $X = (T+1)/4$,*

$$S \leq \frac{4}{3}(3 + \log X) \sum_{\substack{sr^2 \leq X \\ s \text{ squarefree}}} \frac{\tau(4sr^2 + 1)}{sr}$$

$$\leq 2(3 + \mathcal{L})(1 + \mathcal{L})^2 \tau^*(T+2) = \exp\left(O\left(\frac{\log T}{\log \log T}\right)\right).$$

Proof. Clearly $S \leq \sum 1/(\ell-1)$ over triples (m, ℓ, D) as in the definition of F_ℓ ; put $A = A_{m\ell}$. Since $4A \equiv 1$ and $4D \equiv -1 \pmod{m}$, we have $4A^2D^{-1} \equiv -1 \pmod{m}$, so $D \mapsto A^2/D$ preserves $m \mid 4D+1$ and exchanges $D > A$ with $D < A$; hence it suffices to count $D \leq A$ and double. Write $D = sr^2$ with s squarefree. Then $D \mid A^2$ iff $sr \mid A$; write $A = srk$, so that $D \leq A$ iff $k \geq r$. The map $(m, \ell, D) \mapsto (m, s, r, k)$ is injective, and $1/(\ell-1) \leq 2/\ell = 2m/(4A-1) \leq 2m/(3A)$.

From $m \mid 4sr^2 + 1$ and $m \mid m\ell = 4srk - 1$ we get $m \mid 4sr(r+k)$, and $\gcd(m, 4sr) = 1$, so

$$m \mid r+k.$$

Since $k \geq r \geq 1$, the least admissible k is at least $\max(r, m-r) \geq m/2$, so $\sum_{k \equiv -r(m), k \leq X} 1/k \leq (3 + \log X)/m$. Summing over $m \mid 4sr^2 + 1$ gives the first bound; the second follows from $\sum_{s \leq X} 1/s \cdot \sum_{r \leq \sqrt{X}} 1/r \leq (1 + \mathcal{L})^2$ and $4sr^2 + 1 \leq T+2$. $\square$

Remark 3.4. Without the congruence $m \mid r + k$ the same count gives $S \leq T^{1/2-\eta+o(1)}$ for $y = T^{1/2+\eta}$, and the transfer below then only yields $W(p) \gg \log p$. Numerically $S \approx 6.5, 13.6, 23.9, 38.5$ for $T = 10^3, \dots, 10^6$ (with $y = \sqrt{T}$), i.e. S is polylogarithmic in practice.

4. THE ANALYTIC INPUT

Theorem 4.1 (Thorner–Zaman [12, Cor. 1.4 and Rem. 1.5]; cited). *There are absolute effectively computable constants $c_4 > 0$, C_0 such that for coprime a, q with $q \geq 2$ and $x \geq q^{12}$,*

$$\sum_{\substack{p \leq x \\ p \equiv a \pmod{q}}} \log p = \frac{\lambda x}{\varphi(q)} (1 + \varepsilon),$$

$$|\varepsilon| \leq C_0 \left(e^{-c_4 \log x / \log q} + e^{-c_4 (\log x)^{3/5} / (\log \log x)^{1/5}} \right),$$

where $\lambda = 1 - \chi_1(a)x^{\beta_1-1}/\beta_1$ if the exceptional zero β_1 for the modulus q exists and $\lambda = 1$ otherwise. Here, following McCurley [8] as quoted in [12, p. 1], $\prod_{\chi \bmod q} L(s, \chi)$ has no zero in $\Re s \geq 1 - 1/(13 \log(q(|\Im s| + 3)))$ apart from at most one real zero β_1 , which is simple and belongs to a unique real character χ_1 ; also $0 < \lambda < 2$.

We used the arXiv version; we did not compare the numbering with the published version. We also use the quoted McCurley statement for moduli other than those of Theorem 4.1; alternatively the classical Landau–Page theorem [3, Ch. 14] with a constant $c \leq 1/13$ in place of $1/13$ works verbatim.

Lemma 4.2 (One severe character per family; proved modulo the McCurley statement). *Let $\mathcal{Q}$ be a finite set of moduli, all $\leq Z$. Call (χ^*, β) severe if χ^* is a real primitive character whose conductor q^* divides some $q \in \mathcal{Q}$, $L(\beta, \chi^*) = 0$ and $\beta \geq 1 - 1/(13 \log(3Z^2))$. Then there is at most one severe pair. If (χ^*, β^*) is severe and $q^* \mid q \in \mathcal{Q}$, then $\beta_1 = \beta^*$ and χ_1 is induced by χ^* for the modulus q . If no severe pair has $q^* \mid q$, then any exceptional β_1 of q satisfies $x^{\beta_1-1}/\beta_1 \leq 2 \exp(-\log x/(13 \log 3Z^2))$.*

Proof. If $(\chi_a, \beta_a), (\chi_b, \beta_b)$ are severe with conductors dividing $q_a, q_b \in \mathcal{Q}$, put $q' = \text{lcm}(q_a, q_b) \leq Z^2$. Both characters induce characters mod q' with the same zeros in $\Re s > 0$, and both β lie in McCurley’s region for q' ; uniqueness there gives $\beta_a = \beta_b$ and $\chi_a = \chi_b$. The second claim holds because β^* lies in McCurley’s region for $q \leq Z$. For the third, an exceptional zero of q in $[1 - 1/(13 \log 3Z^2), 1)$ would be severe; also $\beta_1 \geq 1 - 1/(13 \log 6) > 1/2$. $\square$

5. A TRANSFER THEOREM

Theorem 5.1 (Proved modulo Theorem 4.1). *Let $T \geq 1$, $Q \geq 2$, and let $B(n) = \sum_{i \in I} c_i \mathbf{1}[n \equiv b_i \pmod{d_i}]$ be a finite real combination with $\gcd(d_i, Q) = \gcd(b_i, d_i) = 1$ and $B(n) \leq \mathbf{1}[W(n) > T]$ for all $n \equiv 1 \pmod{Q}$. Put $Z = Q \max d_i$, $\mu = \sum c_i / \varphi(d_i)$, $M_1 = \sum |c_i| / \varphi(d_i)$, and assume $\mu > 0$ and*

the twist condition: for every real primitive character ψ of conductor $f > 1$ with $\gcd(f, Q) = 1$ and $f \mid d_i$ for some i ,

$$|\mu_\psi| \leq \mu/4, \quad \mu_\psi = \sum_{i: f \mid d_i} c_i \psi(b_i) / \varphi(d_i).$$

Let $K = 1 + \log(M_1/\mu)$. There is an absolute C_1 such that if $\log x \geq C_1 K \max(\log Z, K)$, then some prime $p \leq x$, $p \equiv 1 \pmod{Q}$, has $W(p) > T$.

Proof. Let $q_i = Qd_i$ and let a_i be the class $\equiv 1 \pmod{Q}$, $\equiv b_i \pmod{d_i}$. Then $\sum_{p \leq x, p \equiv 1 \pmod{Q}} \log p B(p) = \sum_i c_i \theta(x; q_i, a_i)$, and $x \geq Z^{12} \geq q_i^{12}$. Put $E_1 = e^{-c_4 \log x / \log Z} + e^{-c_4 (\log x)^{3/5} / (\log \log x)^{1/5}}$ and $E_2 = e^{-\log x / (13 \log 3 Z^2)}$, and apply Lemma 4.2 to $\{q_i\}$. Write the conductor of a severe χ^* as $q^* = q_Q^* q^{*'}$ with q_Q^* supported on the primes of Q and $\gcd(q^{*'}, Q) = 1$, and $\chi^* = \chi_Q^* \chi^{*'}$.

Case A: a severe pair exists with $q^ \mid Q$.* Then for every i , $\lambda_i = 1 - \chi^*(a_i) x^{\beta^*-1} / \beta^* = 1 - x^{\beta^*-1} / \beta^* =: \lambda_* > 0$, because $a_i \equiv 1 \pmod{q^*}$; so $\sum c_i \theta \geq (x \lambda_* / \varphi(Q)) (\mu - C_0 E_1 M_1)$.

Case B: otherwise. If $q^* \mid q_i$ (equivalently $q^{*'} \mid d_i$, $q^{*'} > 1$), then $\lambda_i = 1 - \psi(b_i) x^{\beta^*-1} / \beta^*$ with $\psi = \chi^{*'}$; otherwise $|\lambda_i - 1| \leq 2E_2$. Since $\lambda_i < 2$ and $x^{\beta^*-1} / \beta^* \leq 2$,

$$\begin{aligned} \sum c_i \theta &\geq \frac{x}{\varphi(Q)} (\mu - 2|\mu_\psi| - M_1(2C_0 E_1 + 2E_2)) \\ &\geq \frac{x}{\varphi(Q)} (\mu/2 - M_1(2C_0 E_1 + 2E_2)). \end{aligned}$$

Both brackets are positive once $E_1, E_2 \leq \mu / (8(C_0 + 1)M_1)$, which holds under the stated condition (use $u^{3/5} / (\log u)^{1/5} \geq u^{1/2}$). Hence $\sum_p \log p \mathbf{1}[W(p) > T] \geq \sum_p \log p B(p) > 0$. $\square$

No lower bound for $1 - \beta^*$ is used: in Case A the exceptional factor λ_* multiplies main term and error alike. A prime number theorem valid only for $q \leq \exp(c\sqrt{\log x})$ would force $\log x \gg (\log Z)^2 \geq T$; the Linnik-range uniformity is essential.

6. PROOF OF THEOREM 1.1

Let J be the least even integer $\geq 22S + 10$ and

$$\begin{aligned} B(n) &= \sum_{P \subseteq \mathcal{U}, |P| \leq J-1} (-1)^{|P|} \prod_{\ell \in P} \mathbf{1}[n \bmod \ell \in F_\ell] = \sum_{j=0}^{J-1} (-1)^j \binom{N(n)}{j}, \\ N(n) &= \#\{\ell \in \mathcal{U} : n \bmod \ell \in F_\ell\}. \end{aligned}$$

Pointwise bound. For $N \geq 1$, $\sum_{j \leq J-1} (-1)^j \binom{N}{j} = (-1)^{J-1} \binom{N-1}{J-1} \leq 0$; hence $B \leq \mathbf{1}[N = 0] \leq \mathbf{1}[W > T]$ on $n \equiv 1 \pmod{Q}$ by Lemma 3.1. Expanding $\mathbf{1}[n \bmod \ell \in F_\ell] = \sum_{a \in F_\ell} \mathbf{1}[n \equiv a \pmod{\ell}]$ puts B in the form of Theorem 5.1, with squarefree moduli $d_P = \prod_{\ell \in P} \ell \leq T^J$ coprime to Q .

Mean and mass. With e_j the elementary symmetric functions of the $g(\ell)$, $\mu = \sum_{j < J} (-1)^j e_j$ and $M_1 = \sum_{j < J} e_j \leq e^S$. Let N be a sum of

independent Bernoulli($g(\ell)$) variables; the identity above and $\binom{N-1}{J-1} \leq \binom{N}{J}$ give $|\mu - V| \leq e_J \leq (eS/J)^J \leq e^{-2J}$, where $V = \prod(1 - g(\ell)) \geq e^{-2S}$ (Lemma 3.2). So $\mu \geq 0.99V$ and $\log(M_1/\mu) \leq 3S + 1$.

Twist. If ψ is real primitive of conductor $f > 1$, coprime to Q , dividing some d_P , then $f = \prod_{\ell \in P_f} \ell$ with $r = |P_f| \geq 1$ and $\psi = (\cdot/f)$. With $h(\ell) = \sum_{a \in F_\ell} (a/\ell)/(\ell - 1)$, $|h| \leq g$,

$$\mu_\psi = (-1)^r \prod_{\ell \in P_f} h(\ell) \sum_{P' \subseteq \mathcal{U} \setminus P_f, |P'| \leq J-1-r} (-1)^{|P'|} \prod_{P'} g,$$

which vanishes if $r \geq J$. The inner sum is $V' + \varepsilon'$ with $V' \leq (16/15)^r V$ and $|\varepsilon'| \leq S^{J-r}/(J-r)!$, so $|\mu_\psi| \leq V/15 + e^{-15S} \leq \mu/4$ (treat $r \leq J/2$ and $r > J/2$ separately, using $g \leq 1/16$ and $(e/11)^{1/2} < e^{-0.69}$).

Moduli. $\log Z \leq \log Q + J\mathcal{L} \leq 3y$ because $S = T^{o(1)}$ (Lemma 3.3). Theorem 5.1 gives a prime $p \equiv 1 \pmod{Q}$ with $W(p) > T$ and $\log p \leq C_1(3S+2) \cdot 3y \leq \sqrt{T} \exp((3+\log 2+o(1))\mathcal{L}/\log \mathcal{L})$. Since $840 \mid Q$ the prime is hard, and $p > Q > T$ gives infinitely many distinct primes. Inverting, $\mathcal{L} \geq 2 \log \log p - O(\log \log p / \log \log \log p)$. All inputs are effective. $\square$

7. LIMITS OF THE METHOD

7.1. Complete certificates. A *complete certificate* for $W > T$ is a reduced class $c \pmod{Q}$ all of whose primes have $W(p) > T$. The class of one modulo $L^*(T) = \text{lcm}(24, M \leq T, M \equiv 3 \pmod{4})$ is one, with $\log L^*(T) = (2/3 + o(1))T$.

Proposition 7.1 (Proved). *Every complete certificate has $\log Q \geq (2/3 - o(1))T$.*

Proof. (i) If a prime $\ell \equiv 3 \pmod{4}$, $\ell \leq T$, does not divide Q , then c is compatible with the atom $-4 \in \mathcal{R}(\ell)$ ($D = 1$), and Dirichlet's theorem gives primes with $W \leq \ell$; so $\prod_{\ell \leq T, \ell \equiv 3(4)} \ell \mid Q$. (ii) Since $\mathcal{R}(3) = \{2\}$, $3 \mid Q$ and $c \equiv 1 \pmod{3}$. (iii) Let $\ell \equiv 1 \pmod{4}$, $\ell \leq T/3$, and suppose a prime $r \equiv 2 \pmod{3}$ divides $A = (3\ell + 1)/4$. The atom $M = 3\ell$, $D = r$ has class $-4r \equiv 1 \pmod{3}$; if $\ell \nmid Q$ it is compatible with c , a contradiction. (iv) The primes $\ell \equiv 1 \pmod{4}$ for which $(3\ell + 1)/4$ has no prime factor $\equiv 2 \pmod{3}$ avoid the classes $\ell \equiv -3^{-1} \pmod{r}$ for all primes $r \equiv 2 \pmod{3}$, $r \geq 5$; by the prime number theorem in progressions their logarithmic weight up to X is $o(X)$. Hence $\log Q \geq T/2 + T/6 - o(T)$. $\square$

With Chang's theorem this gives the previous record $W \geq (5/8 - \varepsilon) \log p$, and no complete certificate can do better than linear.

7.2. Prime-local designs. Call a design *prime-local* if it imposes $n \equiv 1 \pmod{\ell^{e_\ell}}$ for ℓ in a set Π and every $M \leq T$, $M \equiv 3 \pmod{4}$, has at most one prime factor outside Π . Theorem 1.1 uses such a design.

Proposition 7.2 (Proved). *In a prime-local design, Π contains every odd prime $\leq \sqrt{T/3}$ with at most one exception. Hence every prime it produces has $\log p \geq (1/\sqrt{3} - o(1))\sqrt{T}$.*

Proof. For odd primes $\ell_1 < \ell_2 \leq \sqrt{T/3}$ outside Π , one of $\ell_1\ell_2, 3\ell_1\ell_2$ is $\equiv 3 \pmod{4}$ and $\leq T$. $\square$

7.3. What would suffice beyond exponent two.

Hypothesis 7.3 ($H_{\min}(\theta)$; open for $\theta < 1/2$). For every $\varepsilon > 0$ and large T there are Q with $840 \mid Q$, $\log Q \leq T^{\theta+\varepsilon}$, and a minorant B as in Theorem 5.1 with $\log \max d_i \leq T^{\theta+\varepsilon}$, $\log(M_1/\mu) \leq T^\varepsilon$, $\mu > 0$, and the twist condition.

Theorem 7.4 (Proved modulo Theorem 4.1). $H_{\min}(\theta)$ implies $W(p) > (\log p)^{1/\theta-\varepsilon}$ for infinitely many hard primes, for every $\varepsilon > 0$.

Proof. Adjoin to Q a prime $\ell_0 \in (R, 2R]$, $R = \max(T, \max d_i)$; this keeps all hypotheses and forces $p > T$. Theorem 5.1 gives $\log p \leq T^{\theta+2\varepsilon+o(1)}$. $\square$

$H_{\min}$ is a purely combinatorial statement about congruences. For $\theta < 1/2$ the residual atoms involve several free primes, and plain event-level Bonferroni truncation over the raw atom list fails: configurations in which $n \equiv -4$ modulo many free primes make $\binom{N-1}{J-1}$ large on a set of measure $\geq T^{-O(\sqrt{J})}$, so the truncated mean is negative for $4.2\theta^2(\log T)^2 \lesssim J \leq T^\theta$. A hub-adapted (hypergraph) minorant would be needed.

8. THE SLICE PARAMETER

For a prime p let $\mathcal{B}_p$ be the set of pairs (c, k) with $1 \leq k \leq \lfloor 2p/3 \rfloor$, $1 \leq c \leq \lfloor (2p+k)/(4k) \rfloor$, $(p, ck) = 1$; put $h = 4ck$, $N_{c,k} = p^2 + 4ck^2$, $s = \text{sf}(c)$, and let $\chi_s = (\Delta_s/\cdot)$, where $\Delta_s < 0$ is the discriminant of $\mathbb{Q}(\sqrt{-s})$; its conductor divides $4s \mid h$. $M_{c,k}(p)$ is the number of divisors of $N_{c,k}$ in the class $-p \pmod{h}$; a divisor e in that class gives the Type I solution $a = (p+e)/h$, $b = (p+N/e)/h$.

Lemma 8.1 (Genus obstruction; proved). *If $p \equiv 1 \pmod{24}$, $(c, k) \in \mathcal{B}_p$ and $\chi_s(p) = 1$, then $M_{c,k}(p) = 0$.*

Proof. Write $c = st^2$. Every prime $q \mid N$ is odd, prime to stk , and $(p(2tk)^{-1})^2 \equiv -s \pmod{q}$, so $\chi_s(q) = 1$; hence $\chi_s(e) = 1$ for every divisor e of N . But $\chi_s(-p) = \chi_s(-1)\chi_s(p) = -1$, and χ_s is defined modulo h . $\square$

Lemma 8.2 (A sufficient prime factor; proved). *If a prime q divides $N_{c,k}$ and $q \equiv -p \pmod{h}$, then $M_{c,k}(p) \geq 1$.*

Proof. q itself is a divisor in the class $-p$. $\square$

Lemma 8.3 (Proved). *Let $p \equiv 1 \pmod{24}$ and suppose $(\ell/p) = 1$ for every prime $5 \leq \ell \leq B$. Then $ck_{\min}(p) > B$. In particular $ck_{\min}(p) \geq n_p$, the least quadratic non-residue modulo p .*

Proof. For $ck \leq B$, $s \leq B$ and $\chi_s(p) = (-s/p) = (-1/p) \prod_{q \mid s} (q/p) = 1$, since $(-1/p) = (2/p) = (3/p) = 1$. Apply Lemma 8.1. $\square$

Proposition 8.4 (Proved). *Let $24 \mid L$ and let $a \bmod L$ be a reduced class with $a \equiv 1 \pmod{24}$ such that every sufficiently large prime $p \equiv a \pmod{L}$ has $ck_{\min}(p) > T$. Then every prime $5 \leq \ell \leq T$ divides L and $(a/\ell) = 1$.*

Proof. Suppose $\ell \nmid L$ or $(a/\ell) = -1$. Choose $c_0 \bmod 4\ell$ with $c_0 \equiv a \pmod{\gcd(L, 4\ell)}$, $c_0 \equiv 1 \pmod{4}$, $(c_0/\ell) = -1$; for $n \equiv 1 \pmod{4}$, $\chi_\ell(n) = (n/\ell)$, so $\chi_\ell(c_0) = -1$. Take a prime $q \equiv -c_0 \pmod{4\ell}$, $q \nmid L$. Then $\chi_\ell(q) = \chi_\ell(-1)\chi_\ell(c_0) = 1$, so $-4\ell \equiv \rho^2 \pmod{q}$ for some $\rho \neq 0$. The conditions $p \equiv a \pmod{L}$, $p \equiv c_0 \pmod{4\ell}$, $p \equiv \rho \pmod{q}$ are compatible and define a reduced class, which contains infinitely many primes. For these, $q \mid p^2 + 4\ell$ and $q \equiv -p \pmod{4\ell}$, so $M_{\ell,1}(p) \geq 1$ (Lemma 8.2), and $(\ell, 1) \in \mathcal{B}_p$ for large p , whence $ck_{\min}(p) \leq \ell \leq T$. $\square$

Corollary 8.5 (Proved). *Let $B(n)$ be a finite combination of residue classes with $B(p) \leq \mathbf{1}[ck_{\min}(p) > T]$ for all large primes p in a reduced class $\equiv 1 \pmod{24}$. Then every large prime p in that class with $B(p) > 0$ has $n_p > T$. Together with Lemma 8.3: the depth certifiable by congruences is exactly n_p .*

Proof. B is periodic modulo some L ; if $B(p) > 0$ then B is positive on the whole class $p \bmod L$, so Proposition 8.4 gives $(\ell/p) = 1$ for $5 \leq \ell \leq T$, and $(2/p) = (3/p) = 1$. $\square$

Theorem 8.6 (Proved modulo [7, Prop. 5.1]). $ck_{\min}(p) \gg \log p \cdot \log \log \log p$ for infinitely many hard primes p .

Proof. Lau and Wu [7, Prop. 5.1] (following Graham–Ringrose [5]) show: for small fixed $\delta > 0$ and $y(x) = \delta \log x \log_3 x$ there are $x_n \rightarrow \infty$ and primes $p \in (x_n^{1/2}, x_n \log x_n]$ with $p \equiv 1 \pmod{4}$ and $(q/p) = 1$ for all primes $q \leq y(x_n)$. Such p are $\equiv 1 \pmod{24}$ and, since $(5/p) = (7/p) = 1$, lie in a Mordell class. Lemma 8.3 gives $ck_{\min}(p) > y(x_n) \geq (\delta/2) \log p \log_3 p$. Effectivity is not claimed. $\square$

Under GRH, Montgomery’s $n_p = \Omega(\log p \log \log p)$ would give the corresponding bound, and Ankeny’s $n_p \ll (\log p)^2$ caps what congruences can certify; the random model predicts $\max_{p \leq x} n_p \asymp \log x \log \log x$ (statements as quoted in [7]). Finally, the primes of Theorem 1.1 are quadratic residues of every $\ell \leq y$, so Lemma 8.3 gives, *jointly*, $W(p) \geq (\log p)^{2-o(1)}$ and $ck_{\min}(p) \geq (\log p)^{1-o(1)}$ for infinitely many hard p (proved modulo Theorem 4.1).

9. THE HAAR SIDE

Impose the class of one at all primes $\leq z$ ($z \geq 3$), i.e. $n \equiv 1 \pmod{Q_z}$ with Q_z defined like Q . For $M \leq T$ write $M = mr$ with m the z -smooth and r the z -rough part. If $r > 1$, the surviving atoms of M are the $D \mid A_M^2$ with $m \mid 4D + 1$, each giving the event $n \equiv -4D \pmod{r}$ of conditional Haar probability $1/\varphi(r)$.

Lemma 9.1 (Proved). *If $M \equiv 3 \pmod{4}$, $D \mid A_M^2$ and $M = mr$ with $m \mid 4D + 1$, then $m \leq r^2 + 1$.*

Proof. By the involution of Lemma 3.3 assume $D \leq A$, $D = sr'^2$, $A = sr'k$, $k \geq r'$. As there, $m \mid r' + k \leq 2k$, so $r = (4sr'k - 1)/m \geq 2sr' - 1/(2k)$, i.e. $r \geq 2sr'$, and $m \leq 4sr'^2 + 1 \leq r^2 + 1$. $\square$

Lemma 9.2 (Proved). *For every $3 \leq z \leq T$, the total mass $S_{\text{tot}} = \sum 1/\varphi(r)$ over all surviving atoms is $\exp(O(\log T / \log \log T))$.*

Proof. Use $1/\varphi(r) \ll \log \log T \cdot m/M$ and repeat the proof of Lemma 3.3 with ℓ replaced by r ; every surviving atom has $m \mid \gcd(4D + 1, M)$, hence $m \mid r' + k$. $\square$

(Using [4, Prop. 1.4] on dyadic blocks one gets $S_{\text{tot}} \ll (\log T)^4 \log \log T$; this is not needed below.)

Proof of Theorem 1.3. Fix $\varepsilon > 0$, put $y = T^{1/3+\varepsilon}$ and $z = y$. Every rough part is ℓ , ℓ^2 or $\ell_1 \ell_2$. Use the independent coordinates $X_\ell = n \bmod \ell^{e_\ell}$. Single atoms form forbidden sets G_ℓ of measure g_ℓ , with $1 \notin G_\ell$ (Lemma 2.2) and $\sum g_\ell \leq S_{\text{tot}}$.

Bad primes. $B = \{\ell : g_\ell > 1/4\}$ has $|B| \leq 4S_{\text{tot}} = T^{o(1)}$. Impose also $n \equiv 1 \pmod{\ell^{e_\ell}}$ for $\ell \in B$ (cost $\leq |B| \log T$). Pairs inside B die by Lemma 2.2; a pair atom $(\ell b, a)$ with $b \in B$ becomes the residue $a \bmod \ell$ at ℓ if $a \equiv 1 \pmod{\ell}$ (b) and dies otherwise. The enlarged sets satisfy $g'_\ell \leq 1/4 + |B| \tau_{\max}^2 T / (\ell y (\ell - 1)) \leq 1/2$, with $\tau_{\max}^2 = \max_{A \leq T} \tau(A^2) = T^{o(1)}$, and $\sum (g'_\ell - g_\ell) \leq |B| \tau_{\max}^2 T / y^2 = T^{1/3-2\varepsilon+o(1)}$.

Good pairs. Under the product measure μ' uniform on the allowed residues, a good pair event has $\mu'(E) \leq 4/((\ell_1 - 1)(\ell_2 - 1))$, and the per-prime weight is $w'_\ell \leq 16\tau_{\max}^2 T / (\ell^2 y) \leq T^{-3\varepsilon+o(1)}$. The asymmetric local lemma [1, Lemma 5.1.1] with $x_E = 2\mu'(E)$ gives $\mu'(\text{no good pair event}) \geq \exp(-16S_{\text{tot}})$.

Collecting. With $Q' = Q_y \prod_{b \in B} b^{e_b}$,

$$\begin{aligned} \delta^*(T) &\geq \frac{8}{\varphi(Q')} \prod (1 - g'_\ell) e^{-16S_{\text{tot}}} \\ &\geq \exp(-\pi(y) \log T - |B| \log T - 18S_{\text{tot}} - T^{1/3-2\varepsilon+o(1)}) \\ &= \exp(-T^{1/3+\varepsilon+o(1)}). \end{aligned}$$

$\square$

The exponent $1/3$ reflects the crude bound $N(M) \leq \tau(A^2)$ for multi-prime atoms at a fixed prime. A congruence saving *uniform in a fixed prime* (the per-prime analogue of Lemma 3.3) would, by the local lemma, give $\log(1/\delta^*(T)) \leq (\log T)^{O(1)}$; we do not know how to prove it. None of this transfers to primes: the local lemma does not provide a pointwise minorant.

10. SUMMARY OF STATUS

- Proved: Lemmas 2.1–3.3, 8.1–8.3, 9.1, 9.2; Propositions 7.1, 7.2, 8.4; Corollary 8.5; Theorem 1.3 (with the classical local lemma and divisor bound).
- Proved modulo Thorner–Zaman [12] (with the McCurley statement quoted there): Lemma 4.2, Theorems 5.1, 1.1, 7.4, and the joint $W/ck_{\min}$ statement.
- Proved modulo Lau–Wu [7, Prop. 5.1]: Theorem 8.6.
- Open: $H_{\min}(\theta)$ for $\theta < 1/2$; a polylogarithmic Haar bound; any bound $W(p) > (\log p)^{2+\delta}$ or $ck_{\min}(p) > (\log p)^{1+\delta}$ infinitely often.

Nothing here bears on the truth of the Erdős–Straus conjecture: a prime with large $W(p)$ or $ck_{\min}(p)$ may well have other representations.

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